

Partisan Pandemic: How Partisanship and Public Health Concerns Affect Individuals' Social Distancing During COVID-19

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Abstract: Rampant partisanship in the United States may be the largest obstacle to the social distancing most experts see as critical to limiting the spread of the COVID-19 pandemic. Analyzing a total of nearly 500,000 responses collected daily between Mar 4 and June 6 reveals not only that partisanship is more important than public health concerns for explaining individuals' social distancing, but also that the effect of partisanship has grown over time – especially among Republicans. All else equal, the relative importance of partisanship for the increasing (un)willingness of Republicans to engage in social distancing highlights the challenge that politics poses for public health.

Main Text:

Public health professionals universally agree that preventing the further spread of the COVID-19 pandemic requires a collective public response (1, 2). Given the increasing level of partisan polarization in the United States (3), the fact that COVID-19 initially impacted Democratic-leaning urban areas more than Republican-leaning rural areas, and differences in how Republican and Democratic political leaders discuss the pandemic, it is unclear whether the pandemic's threat to public health can overcome partisan differences in the perceived severity of the pandemic and the collective need for social distancing to minimize its' impact. Because the effectiveness of social distancing requires widespread compliance to effectively limit the spread of COVID-19, how much do partisan differences affect concerns about COVID-19 and the willingness to social distance?

Decades of research in political science has shown that partisan affiliation is not merely a description of where individuals stand on the ideological spectrum (4), but instead a highly stable (5), emotionally laden (3), social identity (6), that describes many of an individual's social groupings and deeply held values (7). An individual's partisan identity affects the information they collect, process, and respond to (8–11), and the actions they take (12) – including how and whether they choose to receive health care (13–15). Whether partisanship also affects behavior intended to minimize the effects of a global pandemic following widespread scientific agreement is unknown.

We examine the importance of partisan considerations relative to public health concerns arising from the incidence of COVID-19 in local communities on the willingness of individuals to social distance to help limit the spread of COVID-19.

Using daily data on the reported activities of 467,025 U.S. adults collected starting on April 4, 2020, we show that social distancing is 180 times more impacted by partisanship than the incidence of COVID-19 in local communities. Moreover, all else equal, Democrats are 12 percent *more* likely to social distance over time compared to independents, while Republicans have become 24 percent *less* likely to do so. These differences have tremendous consequences for the ability of the United States to limit the spread of COVID-19 and they show the importance of politics and political considerations for public health, even during a global pandemic.

Research Design & Data

Analyzing how partisanship affects social distancing requires individual level data to avoid issues related to ecological inference that may be present when looking at relationships in aggregate units such as counties (16), particularly because Democrats are more likely to live in more population dense places (17) that have been more heavily impacted by COVID-19 to date (18). To track the relationship between partisanship, the incidence of COVID-19 in local communities, and social distancing we interview 467,025 randomly selected respondents from the Survey Monkey platform between April 4, 2020 and June 7, 2020 (19). Every day, an average of 7422 respondents were interviewed and weighted to be representative of the US adult population using the most current estimates from the U.S. Census Bureau's American Community Survey. Weighting ensures that the daily estimates are not only nationally representative, but also directly comparable over time (20, 21).

Our primary outcome measure is whether respondents report going to a restaurant, visiting family or friends, taking a walk, exercising, getting groceries, receiving medical care, and going to work in the last 24 hours. We focus on an additive count of the number of activities an individual reports, but Fig S4 and S5 shows that similar results are obtained when analyzing voluntary activities (e.g., going to a restaurant) and involuntary activities (e.g., seeking medical care) separately.

Relying on any self-report raises concerns about the truthfulness in responses. We feel that a measure which focuses on the last 24 hours is the best solution to mitigate this risk. Focusing on personal activity occurring in the last 24 hours places lower cognitive demand on respondents and it focuses the respondent to think about personal behavior rather than public policy (which may be affected by aspirational political considerations). Asking about support for social distancing measures in general, for example, are more likely to result in responses based on partisan messaging about the severity of the COVID-19 pandemic. Asking about personal behavior in the last 24 hours trades-off an increase in variance (because behavior in the 24 hours may not be representative of their behavior in general) in the hopes of reducing such bias. Our approach is consistent with leading contemporary measures of media exposure that also only ask about the previous 24 hours of activity (23).

To validate our measure, Fig. S3 compares our self-reported mobility index to anonymized cell phone data from Google measuring the change in how long individuals are at various locations (22). The national average of our 24-hour activity measure reassuringly correlates quite strongly (0.89) with Google's measure of change in mobility around retail and recreation.

Our key explanatory variables are the self-reported partisanship of each respondent, and the severity of COVID-19 in their local community. To measure partisanship we rely on the branching question that is standard in the literature (24). Respondents are first asked whether

they identify with the Republican or Democratic Party. Those who identify with neither, are asked whether they “lean” closer to either of the two parties. Only individuals who do not lean are classified as “independent” because “leaners” are known to behave similar to self-identifying partisans(25, 26). We combine leaners with partisans because Table S1 confirms that the behavior of leaners and partisans are largely indistinguishable.

To measure the impact of COVID-19 in each respondent’s community we use the Johns Hopkins Center for Systems Science and Engineering COVID-19 database to calculate the change in COVID-19 cases per 1000 residents in each respondent’s county in the week prior to their interview.¹ Because these numbers have been widely reported, they arguably describe and define respondents’ lived experiences with COVID-19 in their community. Figures S6-S10 show our conclusions are robust to using alternative measures: change in county deaths/1000, absolute county cases/1000, absolute county deaths/1000, change in state cases/1000, and change in state deaths/1000.

Two results are of interest: 1) how much of the variation in social distancing behavior can be attributable to partisanship relative to the impact of COVID-19 in the local community, and 2) the conditional effect of partisanship and COVID-19 community cases on social distancing.

To begin, Figure 1 summarizes the patterns in the weighted data over time to demonstrate how these considerations relate to COVID-19 concerns and social distancing behavior. The top row of plots (A, B) graph the percentage of respondents who report being “very worried” or “somewhat” worried about catching COVID-19 each day between February 11, 2020 and June 6, 2020 using 642,760 interviews. The bottom row (C, D) plots the average number of activities individuals report doing in the last 24 hours using the 467,025 interviews conducted starting on March 4, 2020. The columns in Figure 1 report the daily averages by partisanship (A, C) and the quintile of population adjusted COVID-19 cases in the respondents’ county (B, D).²

Several results immediately emerge from the raw data. First, although Democrats are more concerned with catching COVID-19 than Republicans throughout, the concerns of both were increasing until early April when Republicans began to become less concerned and Democrats maintained their level of concern. By the end of the series, 80.9 % of Democrats were concerned, but only 41.3% of Republicans report being concerned. Second, Panel (B) reveals that these differences are largely unrelated to differences in the incidence of COVID-19 in the respondents’ communities – the average level of concern does not vary depending on whether a respondent lives in one of most or least impacted counties. Figure S11 displays the same trends for different age groups. While Americans over the age of 65 are slightly more likely to socially distance, this difference does not change over time, and likely represents persistent differences in social activities for older individuals.

¹ Survey Monkey asks each respondent for their home zip code. 81% of our respondents reside in a zip codes that is located entirely within a single county. For zip codes in multiple counties we assign the respondent to the county where most zip code residents reside. 97% live in a zip code where at least ¾ of the population live in a single county.

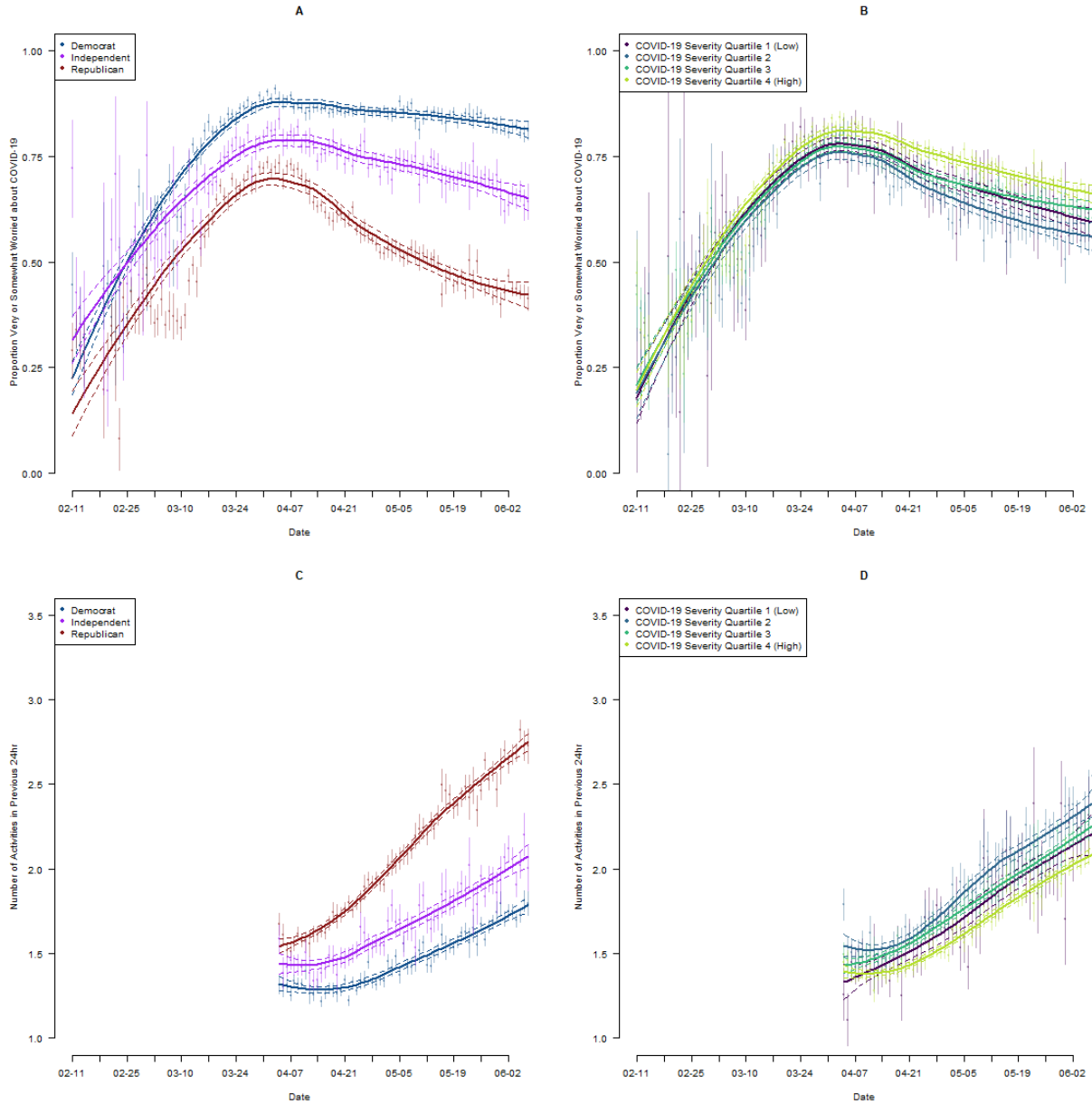


Fig. 1. Concern with Catching COVID-19 and Social Distancing Behavior Over time by Partisanship and COVID-19 Impact in the Community. Plots (A, B) graph the weighted daily percentage of respondents who are “very” or “somewhat” worried about catching COVID-19 and Plots (C, D) graph the average number of social activities individuals report doing in the last 24 hours. Plots (A, C) report the daily averages by partisanship and Plots (B, D) report the average for each quintile of population adjusted COVID-19 cases in respondents’ counties. 95% confidence intervals are reported for each daily average and plotted lines are loess smoothers to summarize trends over time.

The social distancing consequences of the variation in COVID-19 concerns is clear from the patterns over time. When the social distancing questions were originally asked in early April there were few partisan differences – reflecting that Democrats and Republicans were similarly concerned. Over time, plot C shows that although all respondents report engaging in more social activities over time on average, the increase is greatest among Republicans and the average number of social activities increases by 1.05 (a 63% increase) (compared to 0.41 (26% increase) among pure independents and 0.48 (37% increase) among Democrats. Considering the average number of social activities by COVID-19 impact in the local community (Plot D) reveals less social activity in the most impacted counties, but the differences between the most and least impacted counties is considerably less than the differences between Democrats and Republicans.

To decompose the precise impact of partisanship and COVID-19 on the willingness of individuals to social distance we estimate several daily regression models to estimate the conditional effect of each on social distancing behavior as well as how much variation in social distancing behavior is explained by each. We regress the level of self-reported activity in the last 24 hours for individual i in state j on state fixed-effects (α_j) (to control for between-state differences) and a matrix of individual-level and zip-code level demographics (K_i) to account for other possible sources of variation.³ We then estimate models that iteratively add party indicators (with pure independents as the excluded category) and the change in COVID-19 deaths in the individual's county in the previous week. The four models we estimate for each day t are:

Equation 1

$$Last24_i = \alpha_j + \beta K_i + \epsilon_{ij}$$

Equation 2

$$Last24_i = \alpha_j + \gamma Democrat_i + \zeta Republican_i + \beta K_i + \epsilon_{ij}$$

Equation 3

$$Last24_i = \alpha_j + \eta County.Change.in.COVID.Cases_i + \beta K_i + \epsilon_{ij}$$

Equation 4

$$Last24_i = \alpha_j + \gamma Democrat_i + \zeta Republican_i + \eta County.Change.in.COVID.Cases_i + \beta K_i + \epsilon_{ij}$$

The first result of interest is provided by the coefficients of partial determination⁴ for partisanship and the change in COVID-19 cases in the respondent's county. These quantities characterize how much of the residual variation in Equation 1 is explained by adding partisanship (Equation 2) or COVID-19 cases (Equation 3), which summarizes the extent to which variation in social distancing can be explained by partisanship and public health considerations. The conditional marginal

³ Demographic covariates are gender, age, race, education, income, population density, and employment status.

⁴ The coefficient of partial determination is determined by dividing the difference in the residual sum of squares between the demographic and fuller models by the residual sum of squares in the demographic model. In other words, it determines how much of the variation left unexplained by demographics and state fixed-effects is explained by our two key independent variables.

effects of partisanship and COVID-19 cases on social distancing behavior are provided by a regression including both partisanship and COVID-19 considerations (Equation 4).

Results

Figure 2 reports percentage of residual variation in social distancing behavior that is explained by partisanship and the local incidence of COVID-19 over time. In early March, neither partisanship nor public health concerns explain a large amount of the residual variation in distancing behavior – primarily because there was little variation to explain given that most individuals were social distancing. Over time, partisan variation explains an increasing percentage of variation in social distancing; nearly 8% of the variation in distancing behavior that is not explained by demographics or state is explained by the party affiliation of the respondent in May. Public health, on the other hand, consistently explains only a small amount of this residual variation.

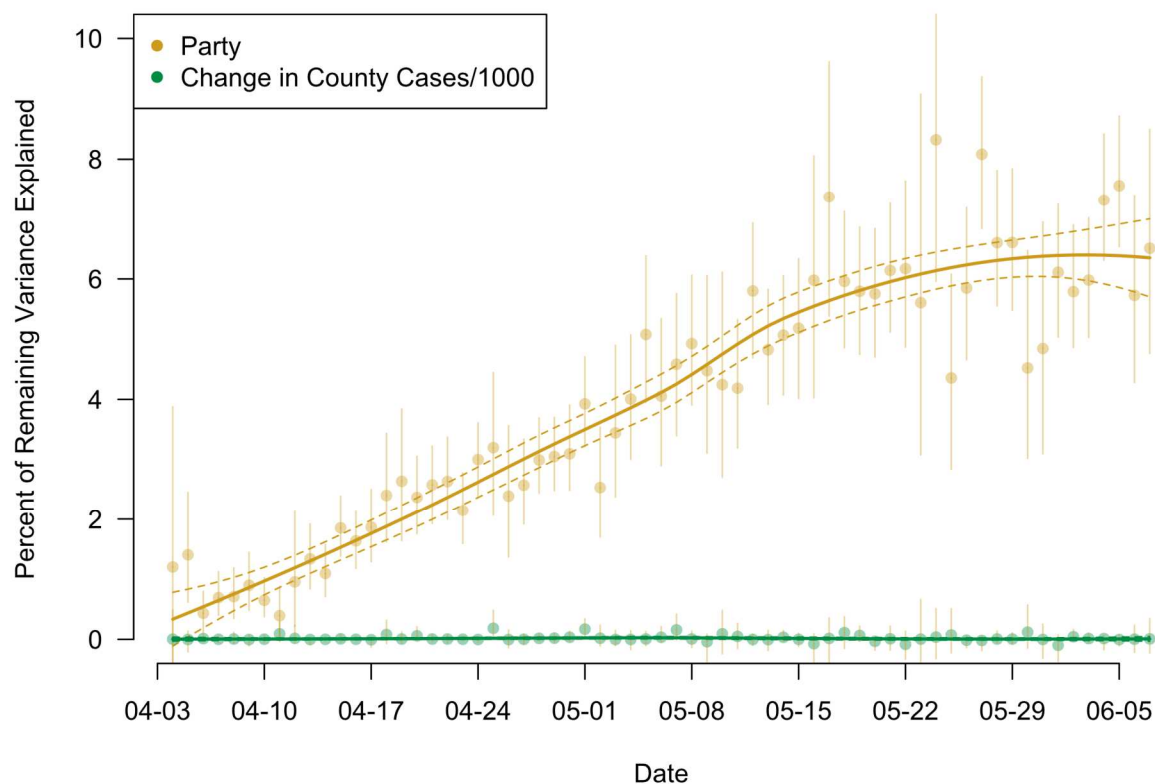


Fig. 2. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community. The daily coefficients of partial determinations are computed via OLS regression using specifications (1-3) run separately on each day. Observations weighted to the US adult population. They describe the percent of residual variation in specification (1) that is explained by our two key variables. Standard errors are from 1000 bootstrap samples for each day, and represent the 95% distribution of bootstrap estimates. Trend is local regression weighted by the inverse of each estimates standard error.

To determine how partisanship and local COVID-19 cases are predicted to impact individuals' social distancing, Figure 3 graphs the marginal effect of partisanship (with independent as the base category) and change in COVID-19 deaths over time from specification (4). Whereas Figure 1 (Plot C) presents the weighted daily averages in the raw data, Figure 3 graphs the marginal effect of each consideration in a model using statistical controls to allow for all-else-equal comparisons.

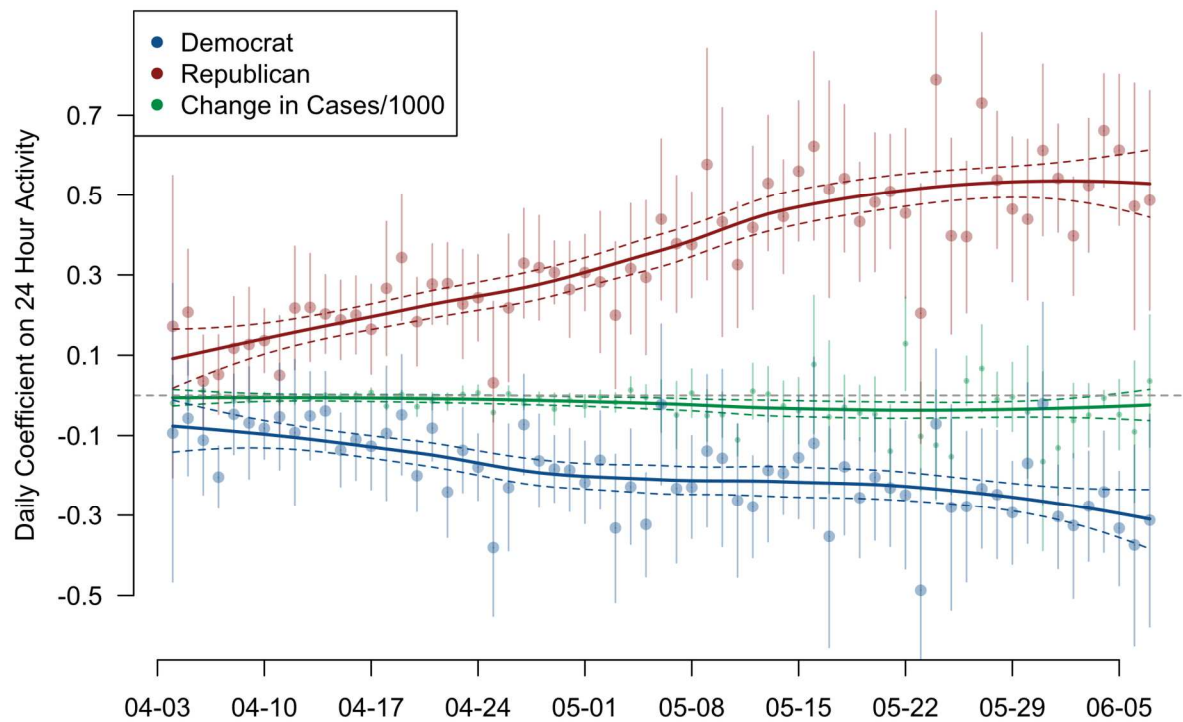


Fig 3. Marginal Conditional Effect of Partisanship and COVID-19 cases on Social Distancing over time. Coefficients are from OLS regression using specification (4) run separately on each day. Observations weighted to the US adult population. For party variables, reference is pure independents. Partisan leaners are included with identifying partisans. 95% confidence intervals from OLS standard errors clustered by state. Trend is local regression weighted by the inverse of each estimates standard error.

The results are striking. The blue line in Figure 3 reveals that Democrats are less likely to engage in social activities than independents and they are increasingly likely to so do over time. By early June Democrats were engaging in -0.31 less activities on average than independents all else equal. In contrast, Republicans (red line) are increasingly more likely to engage in social activities than independents such that they are engaging in more than 0.6 social activities than pure independents on average by the end (red line). Altogether, Republicans are therefore engaging in 0.93 additional social activities on average than Democrats in early June. Moreover,

the increasing effect of partisanship on social distancing is asymmetric: all else equal, Republicans are increasing their social activity far more than other Americans.

While partisanship matters greatly for social distancing, the impact of the population adjusted change in COVID-19 cases in a respondent's county (green line) has little to no statistically distinguishable effects on social distancing. There is a relationship between the amount of social distancing and COVID-19 cases in mid-to-late May, but detectable differences do not persist.

Our statistical controls rule out the most plausible alternative hypotheses for differences that might be related to age, race, education, gender, employment status, or population density. The differences we identify also cannot be explained by Republicans living in states where social activity is more permissible (whether *de jure* or *de facto*) because the effects being identified are based on within-state differences given the state fixed effects we employ.

Looking at partisan differences within each state confirms the importance of partisanship for social distancing behavior regardless of the impact of COVID-19 in the state. Figure 4 plots the conditional marginal effects of partisanship on social distancing for each state using the same statistical controls and a polynomial time trend to account for other possible time-varying influences in the state.

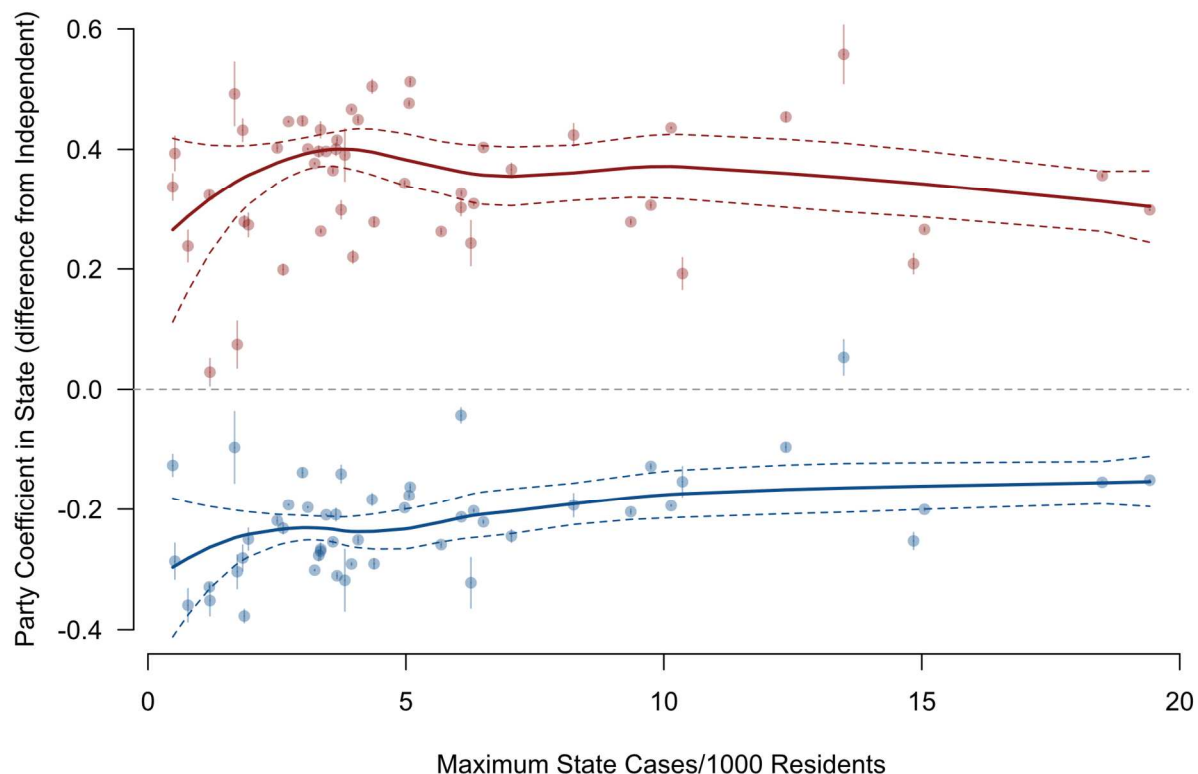


Fig 4. Marginal Conditional Effects of Partisanship by State. The coefficients and 95% confidence intervals are from estimating using OLS from a pooled model of same-state respondents using specification (4). Because we pool all respondents from a state over time, a polynomial time trend is used to account for time-varying differences. Coefficients are in relation to pure independents. The separate state results are reported according to the state's

maximum recorded cases/1000 residents (x-axis) to relate the effects to the incidence of COVID-19 in the state. The between-state trendline is a local area regression weighted by the inverse of each estimates standard error.

For purposes of comparing the variation in the effect of partisanship to the variation in COVID-19 impact between states, we plot the marginal conditional effects Democrat and Republican partisanship against the maximum number of COVID-19 cases rate per 1000 residents. As Figure 4 makes clear, there is essentially no relationship between these sets of values. Partisanship is not more influential in states that are more or less impacted by COVID-19 ; there is a remarkably consistent effect of party in each state⁵, largely conforming to previous research which finds an increasingly nationalized politics in the United States (27–29).

Discussion

Our results point to an unequivocal conclusion: partisanship is a far more important determinant of an individual's response to the COVID-19 pandemic than the impact of COVID-19 in that individual's local community. The implications of this are significant. The COVID-19 pandemic in the United States is currently as much a political problem as it is a public health problem. Given the differences in perceptions and behavior we document between Democrats and Republicans exposed to the same level of COVID-19 in their local communities, simply highlighting the public health concerns associated with COVID-19 seems unlikely to encourage the collective social distancing that is required to help mitigate the further spread of the pandemic. Instead, political leadership – especially by Republicans – seems essential for changing the partisan-related differences we document in individuals' behaviors and opinion.

References and Notes

1. Center for Disease Control, Social Distancing, Quarantine, and Isolation, (available at <https://www.cdc.gov/coronavirus/2019-ncov/prevent-getting-sick/social-distancing.html>).
2. World Health Organization, Advice for public, (available at <https://www.who.int/emergencies/diseases/novel-coronavirus-2019/advice-for-public>).
3. S. Iyengar, S. J. Westwood, Fear and Loathing across Party Lines: New Evidence on Group Polarization. *Am. J. Pol. Sci.* (2015), doi:10.1111/ajps.12152.
4. A. Campbell, P. E. Converse, W. E. Miller, D. E. Stokes, *The American Voter* (University of Chicago Press, 1960).
5. P. E. Converse, The nature of belief systems in mass publics (1964). *Crit. Rev.* (2006), , doi:10.1080/08913810608443650.
6. D. P. Green, B. Palmquist, E. Schickler, *Partisan hearts and minds: Political parties and the social identities of voters* (Yale University Press, 2004).
7. L. Mason, *Uncivil agreement: How politics became our identity* (University of Chicago Press, 2018).
8. J. R. Zaller, *The nature and origins of mass opinion* (Cambridge university press, 1992).

⁵ Only the District of Columbia has results contrary to expectations, where Democrats are slightly *more* likely to be socially active than independents. This being said, it is still the case in DC that Republicans are far more active than either Democrats or independents.

9. G. S. Lenz, *Follow the leader?: how voters respond to politicians' policies and performance* (University of Chicago Press, 2013).
10. M. Levendusky, *How partisan media polarize America* (University of Chicago Press, 2013).
11. C. H. Achen, L. M. Bartels, *Democracy for realists: Why elections do not produce responsive government* (Princeton University Press, 2017), vol. 4.
12. M. F. Margolis, M. W. Sances, Partisan differences in nonpartisan activity: The case of charitable giving. *Polit. Behav.* **39**, 839–864 (2017).
13. M. A. Baum, Red state, blue state, flu state: Media self-selection and partisan gaps in swine flu vaccinations. *J. Health Polit. Policy Law.* **36**, 1021–1059 (2011).
14. A. E. Lerman, M. L. Sadin, S. Trachtman, Policy uptake as political behavior: evidence from the Affordable Care Act. *Am. Polit. Sci. Rev.* **111**, 755–770 (2017).
15. M. W. Sances, J. D. Clinton, Who participated in the ACA? Gains in insurance coverage by political partisanship. *J. Health Polit. Policy Law.* **44**, 349–379 (2019).
16. S. Piantadosi, D. P. Byar, S. B. Green, The ecological fallacy. *Am. J. Epidemiol.* **127**, 893–904 (1988).
17. J. G. Gimpel, N. Lovin, B. Moy, A. Reeves, The Urban–Rural Gulf in American Political Behavior. *Polit. Behav.*, 1–26 (2020).
18. Center for Disease Control, S. Bialek, V. Bowen, N. Chow, A. Curns, R. Gierke, A. Hall, M. Hughes, T. Pilishvili, M. Ritchey, K. Roguski, B. Silk, T. Skoff, P. Sundararaman, E. Ussery, M. Vasser, H. Whitham, J. Wen, “Geographic Differences in COVID-19 Cases, Deaths, and Incidence—United States, February 12–April 7, 2020” (2020).
19. Survey Monkey, Survey methodology: How SurveyMonkey gets its data | SurveyMonkey, (available at <https://www.surveymonkey.com/mp/survey-methodology/>).
20. J. N. Druckman, C. D. Kam, in *Cambridge Handbook of Experimental Political Science* (2012).
21. J. D. Clinton, A. M. Engelhardt, M. J. Trussler, Knockout blows or the status quo? Momentum in the 2016 primaries. *J. Polit.* **81** (2019), doi:10.1086/703383.
22. Google, COVID-19 Community Mobility Reports, (available at https://www.google.com/covid19/mobility/data_documentation.html).
23. S. Dilliplane, S. K. Goldman, D. C. Mutz, Televised Exposure to Politics: New Measures for a Fragmented Media Environment. *Am. J. Pol. Sci.* (2013), doi:10.1111/j.1540-5907.2012.00600.x.
24. The American National Election Studies, The ANES Guide to Public Opinion and Electoral Behavior, (available at <https://electionstudies.org/resources/anes-guide/>).
25. B. E. Keith, E. Orr, M. C. Westlye, R. E. Wolfinger, D. B. Magleby, C. J. Nelson, The Partisan Affinities of Independent ‘Leaners’.’ *Br. J. Polit. Sci.* **16**, 155–185 (1986).
26. S. Klar, Y. Krupnikov, *Independent Politics* (Canbridge University Press, 2016).
27. D. J. Hopkins, *The increasingly United States: How and why American political behavior*

nationalized (University of Chicago Press, 2018).

28. G. C. Jacobson, It's nothing personal: The decline of the incumbency advantage in US House Elections. *J. Polit.* (2015), doi:10.1086/681670.
29. M. Trussler, Get Information or Get in Formation: The Effects of High-Information Environments on Legislative Elections. *Br. J. Polit. Sci.* (2020), doi:10.1017/S0007123419000577.

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Data and materials availability: All data, code, and materials used in the analysis is available upon request to any researcher for purposes of reproducing or extending the analysis.

List of Supplementary Materials:

Section S1. Study design materials and methods

Section S2. Method of Statistical Analysis

Section S3. Robustness of Results: Using Specific Activity Measures instead of Sum Total

Section S4: Robustness of Results: Differences in Leaners and Identified Partisans

Section S5: Robustness of Results: Alternative Measures of COVID-19 Impact

Figure S1: Daily Number of Respondents to “Last 24-hr activity measure.”

Figure S2: Daily Trends in each Social Activity used to create the “Last 24hr activity” measure

Figure S3: Comparison of Statewide Average of Total Self-Reported Activity to Google Mobility Data.

Figure S4. Coefficient plot of Specification (4) by Social Activity.

Figure S5: Effect of Partisanship and COVID-19 on Individual Activities Over Time.

Fig. S6. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (Change in County Deaths/1000).

Fig. S7. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (County Cases/1000).

Fig. S8. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (County Deaths/1000).

Fig. S9. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (Change in State Cases/1000).

Fig. S11. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (Change in State Deaths/1000).

Fig. S10. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (Change in State Deaths/1000).

Table S1. Number of Social Activities Regression Results

Supplementary Materials for

**Partisan Pandemic: How Partisanship and Public Health Concerns Affect
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Materials and Methods

Section S1: Study design materials and methods

To obtain our data, we interview 467,025 randomly selected respondents from the Survey Monkey platform between April 4, 2020 and June 7, 2020. These individuals are randomly sampled from the approximately 2 million individuals who take Survey Monkey surveys every day.

These data are weighted for age, race, sex, education, and geography using the Census Bureau's American Community Survey to reflect the demographic composition of the United States. An additional smoothing parameter for political party identification based on aggregates of SurveyMonkey research surveys is included. Weights are generated each day using daily interviews Monday through Friday, and once over combined interviews on Saturday and Sunday. Party identification parameters are refreshed bi-weekly.

Figure S1 Displays the number of valid responses to our key dependent variable on each day from April 4th to June 7th, 2020.

The following question is asked to determine an individual's degree of social activity, which serves as our main dependent variable.

In the past 24 hours, have you left your home to do any of the following? (Select all that apply)

[Response order randomized]

- Go to work
- Go grocery shopping
- Exercise
- Go for a walk/get some fresh air
- Visit with friends or family
- Eat at a restaurant or bar
- Get medical care
- None of the above

To measure each individual's level of activity in the previous 24-hours, we add up the number of activities they reported participating in. We are cognizant of the fact that this combines activities that are voluntary (like going to a restaurant) and those that are involuntary (like getting medical care). Figure S2 displays the trends in each of these activities over our time, and clearly there is much more variation in more voluntary activities. To probe these differences, in Figure S4 and S5 below we report the partisan differences in each of these activities. While there is notable heterogeneity, in each activity the same pattern of partisan-motivated activity is present.

The following question was asked to gauge an individual's level of worried about COVID-19:

How worried are you that you or someone in your family will be exposed to the coronavirus?

- Very worried
- Somewhat worried
- Not too worried
- Not worried at all

To classify individuals as partisans they are asked the following questions:

In politics today, do you consider yourself a Republican, Democrat, or Independent?

- Republican
- Democrat
- Independent

[If respondent selects “Independent”] *As of today, do you lean more to the Republican or Democratic Party?*

- Republican
- Democrat
- Neither

If respondent answers Republican or Democrat in the first question, or leans towards either of those parties in the second question, they are classified accordingly. Respondents who select “Neither” on the second question are classified as “independent”.

Data on COVID-19 prevalence comes from COVID-19 Data Repository by the Center for Systems Science and Engineering (CSSE) at Johns Hopkins University. These data report, on each day, the number of COVID-19 cases in each county in the United States.

As a first data processing step, we use county population estimates from the American Community Survey to transform county cases to be cases per 1000 residents.

We then calculate for each county for each week the change in the number of cases from the previous week in that county, using the median number of cases in each county for each week.

Consider, for example, interviewing an individual in zip-code 37206 (in Davidson county on Nashville’s East Side) on April 4th. On the week of April 4th in Davidson county (in which the zip-code is wholly contained) the median number of cases per 1000 was 1.001. In the week previous, the median number of cases per 1000 was .428. This individual, therefore, would be given the value .573 for change in county cases per 1000. This value is a fair representation of the fact that this individual, on this day, was witnessing a worsening public health situation in their immediate surrounding.

Survey Monkey respondents self-report their zip-code. To assign respondents to a county we determine the county in which the majority of the residents of that zip-code live in. For 86% of respondents, their zip-code is entirely contained within one county.

Finally, we merge the data on how the number of COVID-19 cases is changing in each county on each day to the Survey Monkey data.

Section S2. Method of Statistical Analysis

Our methodology estimates the following 4 models separately on each day t for individuals i in states j .

Equation 4

$$Last24_i = \alpha_j + \beta K_i + \epsilon_{ij}$$

Equation 5

$$Last24_i = \alpha_j + \gamma Democrat_i + \zeta Republican_i + \beta K_i + \epsilon_{ij}$$

Equation 6

$$Last24_i = \alpha_j + \eta County.Change.in.COVID.Cases_i + \beta K_i + \epsilon_{ij}$$

Equation 4

$$Last24_i = \alpha_j + \gamma Democrat_i + \zeta Republican_i + \eta County.Change.in.COVID.Cases_i + \beta K_i + \epsilon_{ij}$$

On each day, we regress the level of self-reported activity in the last 24 hours for individual i in state j on state fixed-effects (α_j) (to control for between-state differences) and a matrix of individual-level and zip-code level demographics (K_i). These demographics are gender (dummy for female), age (dummies for 18-24, 25-34, 34-44, 45-54, 55-64, 65+), race (dummies for White, Black, Asian, Hispanic, and other), education (dummies for less than high school, high school only, associates or some college, bachelor degree, postgraduate degree) , income (dummies for below 29.9k, 30k-74.9k, 75k-149.9k, +150k), zip code population density (dummies for urban, suburban, and rural), and employment status (dummy for unemployed).

In all models, standard errors are clustered at the state level.

To estimate the coefficients of partial determination for Figure 2, we calculate the following equation using specifications 1, 2, & 3:

$$\delta_{2|3} = (SS_{res2|3} - SS_{res1})/SS_{res1}$$

Thus, the quantity $\delta_{2|3}$ is the proportion of unexplained variance in specification 1 that is explained by adding partisanship (in specification 2) or public health (in specification 3).

To obtain standard errors for these estimates we perform a bootstrapping procedure. On each day we randomly sample (with replacement) rows from the dataset to construct a bootstrap sample the same length as the day's original sample. In this bootstrap sample we estimate specifications 1-3 and record the coefficients of partial determination. We complete this procedure 1000 times per day. The standard errors in Figure 2 represent the 95% distribution of these bootstrap estimates.

Figure 3 represents the daily coefficients on the two-party variables and change in county cases on each day from specification (4). The 95% confidence intervals are derived from the standard errors from the OLS model, which are clustered by state.

The coefficients in Figure 4 come from regressions using specification (4) run separately in each state. For these regressions, individuals are pooled across days, so a polynomial time trend is employed. The values on the x axis are, for each state, the maximum cases/1000 recorded.

Section S3. Robustness of Results: Using Specific Activity Measures instead of Sum Total

Our main dependent variable is an additive index of activities the respondent has participated in over the previous 24-hours. This scale includes both voluntary and involuntary activities. A reasonable critique is that partisanship's effects on this measure is being driven entirely by differential activity in voluntary activity.

To investigate this we repeat our main analysis separately for each activity.

Figure S4 presents results from specification (4) run separately on each activity, pooled across time with a polynomial time trend. While there are important differences, in each activity partisanship is a significant differentiator for activity. As with the main results in the paper, these results cannot be explained by differences between states (as the models include state fixed effects), nor can they be explained by an individual's age, race, income, education, unemployment status, or population density of their community.

Figure S5 presents results from running specification (4) on each activity on each day, to determine the trends in the impact of partisanship. Interestingly, some activities (like going to work or walking) have seen persistent partisan differences across time. Others seem to be far more affected by time. Going to a restaurant, for instance, has seen a more recent explosion in partisan differences.

Section S4. Robustness of Results: Differences in Leaners and Identified Partisans

In Table S1 we present a pooled model using specification (4) where party "leaners" are separated. As mentioned in the main text, political science literature has consistently found that leaners behave in ways broadly similar to identified partisans. The results in this case support that conclusion. Democrat leaners and identified Democrats are similarly less likely to be socially active compared to Independents, and Republican leaners and identified Republicans are similarly more likely to be socially active compared to Independents. Indeed, in the case of levels of activity during this pandemic, being a leaner is *more* impactful than being an identified partisan.

Section S5. Robustness of Results: Alternative Measures of COVID-19 Impact

For our main analyses we used as a measure of public health the weekly change in the number of cases per 1000 in a respondent's county. We believe that this is a reasonable measure of how an individual's immediate public health situation is changing. However, to show the robustness of our findings in this section we replicate Figure 2 in the paper using various other measures of

5 public health: the weekly change in the number of deaths in an individual's county, the total number of cases in an individual's county, the total number of deaths in an individual's county, the weekly change in the number of cases in an individual's state, and the weekly change in the number of deaths in an individual's state. Note that for these last two measures we drop the state fixed effects from the specification, as otherwise the model would not be able to be estimated.

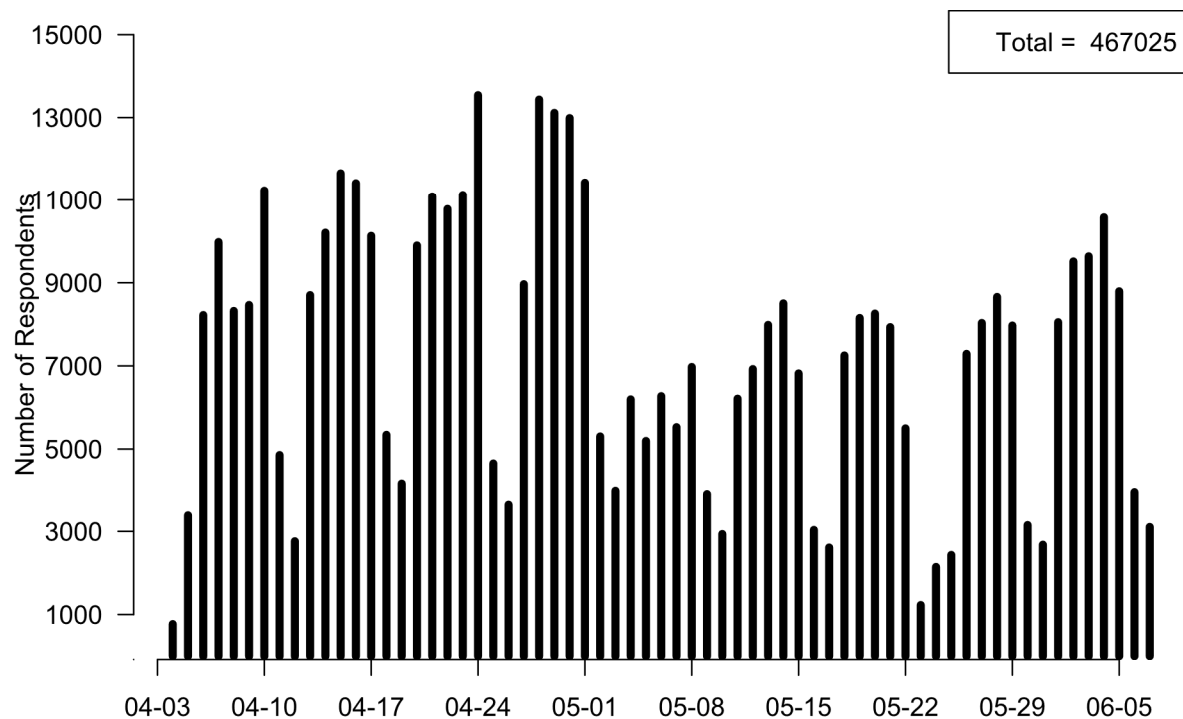


Figure S1: Daily Number of Respondents to “Last 24-hr activity measure.” The number of responses fluctuates daily due to the differences in the number of respondents who were invite to participate each day because of demands for other surveys as well as in the timing of when respondents choose to respond to a request to take a survey. A total of 467,025 responses were analyzed.

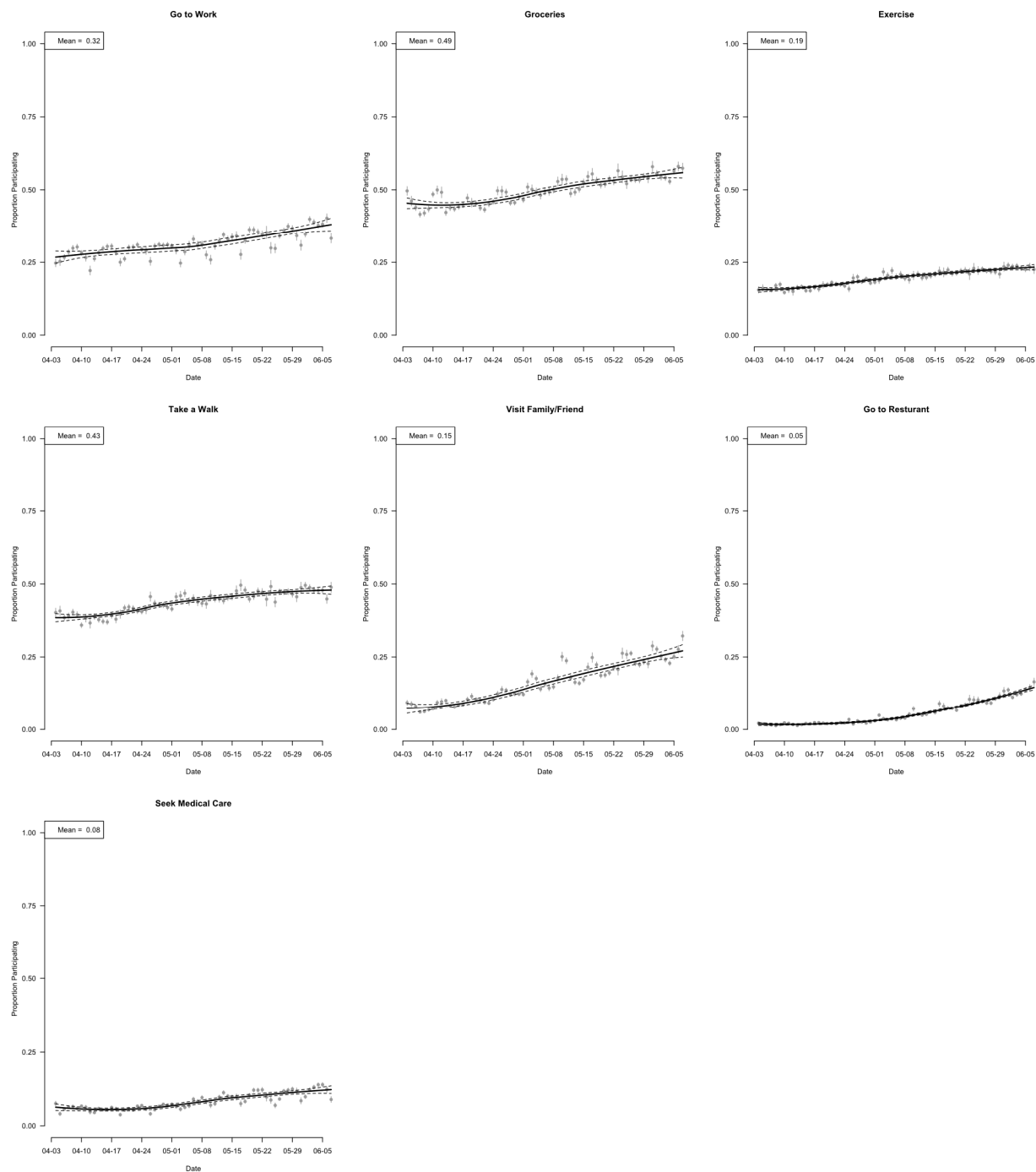


Figure S2: Daily Trends in each Social Activity used to create the “Last 24hr activity” measure

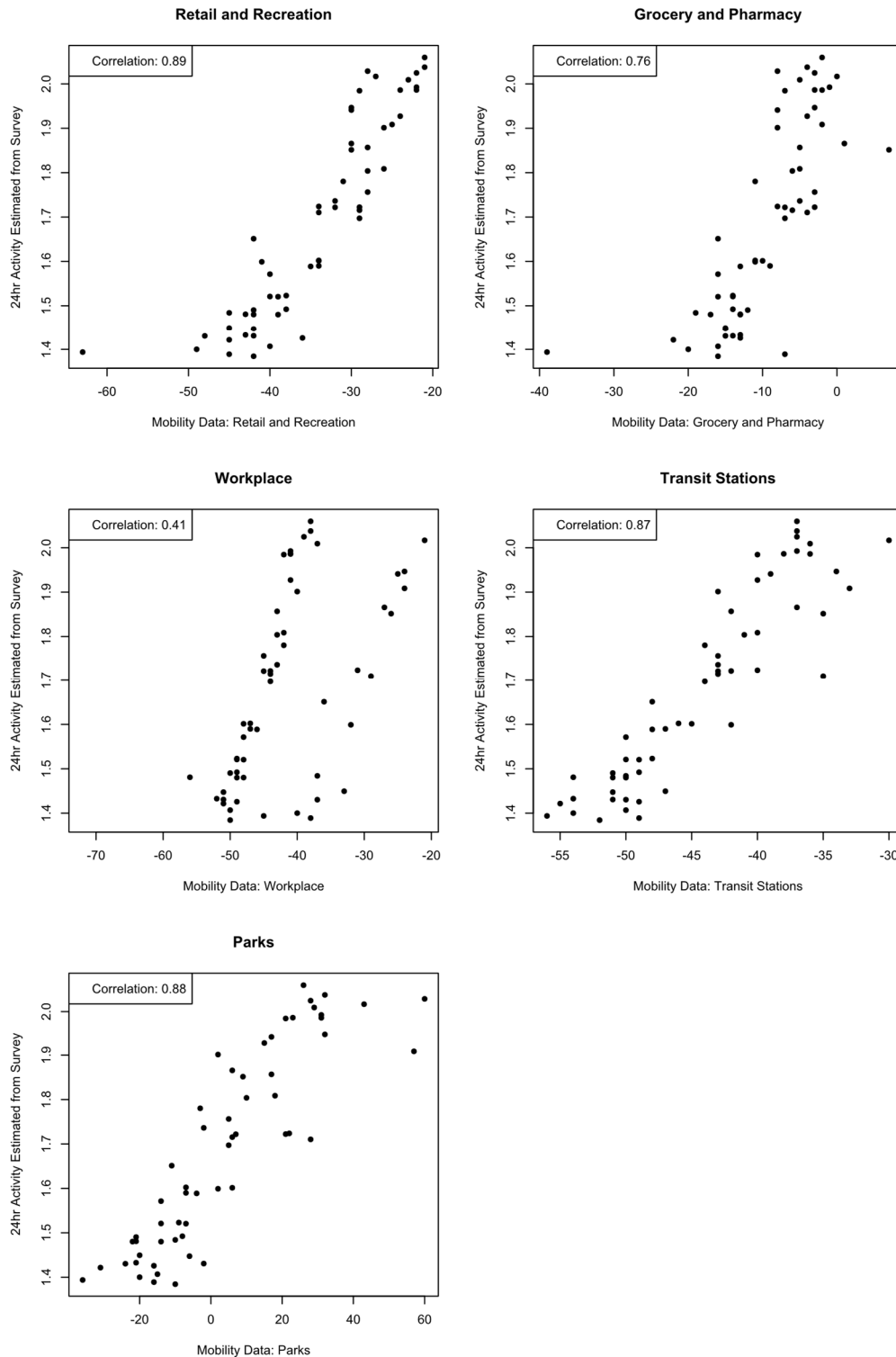


Figure S3: Comparison of Statewide Average of Total Self-Reported Activity to Google Mobility Data. Y-axis plots the average number of social activities by state pooled over time, x-axis plots Google's mobility measure for various geographic locations.

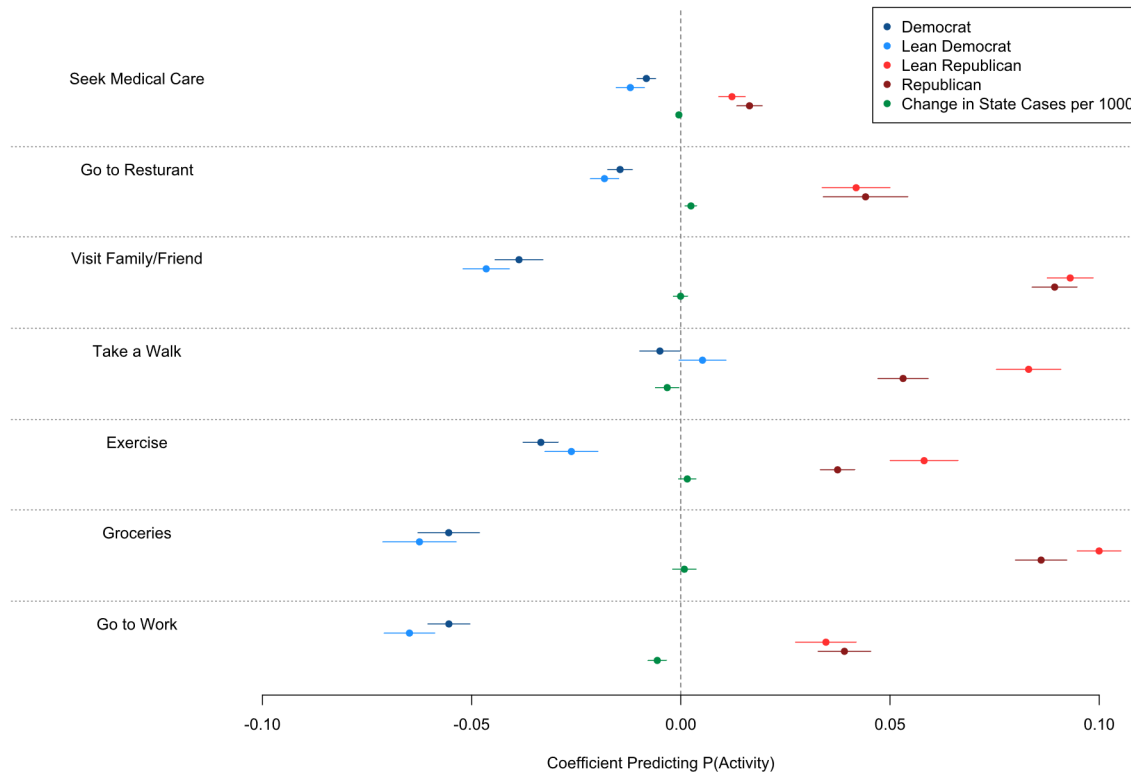


Figure S4. Coefficient plot of Specification (4) by Social Activity. Results predict the probability of engaging in each of the activities by partisanship and COVID-19 incidence to show robustness of partisan differences across activities. Each row reflects the results from a separate pooled regression, which includes a polynomial time trend, state fixed effects, and a battery of demographics. Standard errors are clustered by state.

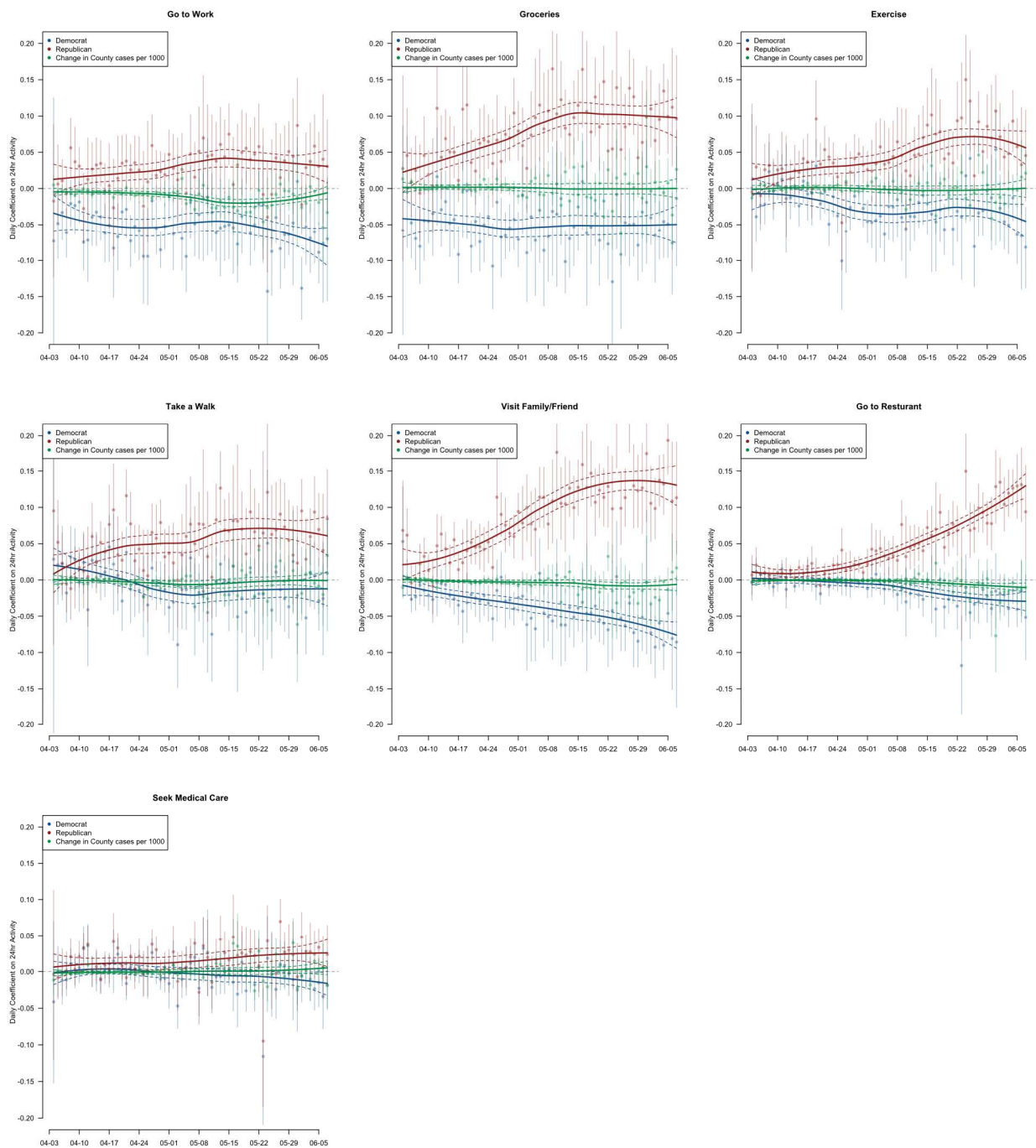


Figure S5: Effect of Partisanship and COVID-19 on Individual Activities Over Time. Coefficients are from OLS regression using specification (4) run separately on each day. Observations weighted to the US adult population. For party variables, reference is pure independents. Partisan leaners are included with identifying partisans. 95% confidence intervals from OLS standard errors clustered by state. Trend is local regression weighted by inverse of each estimates standard error.

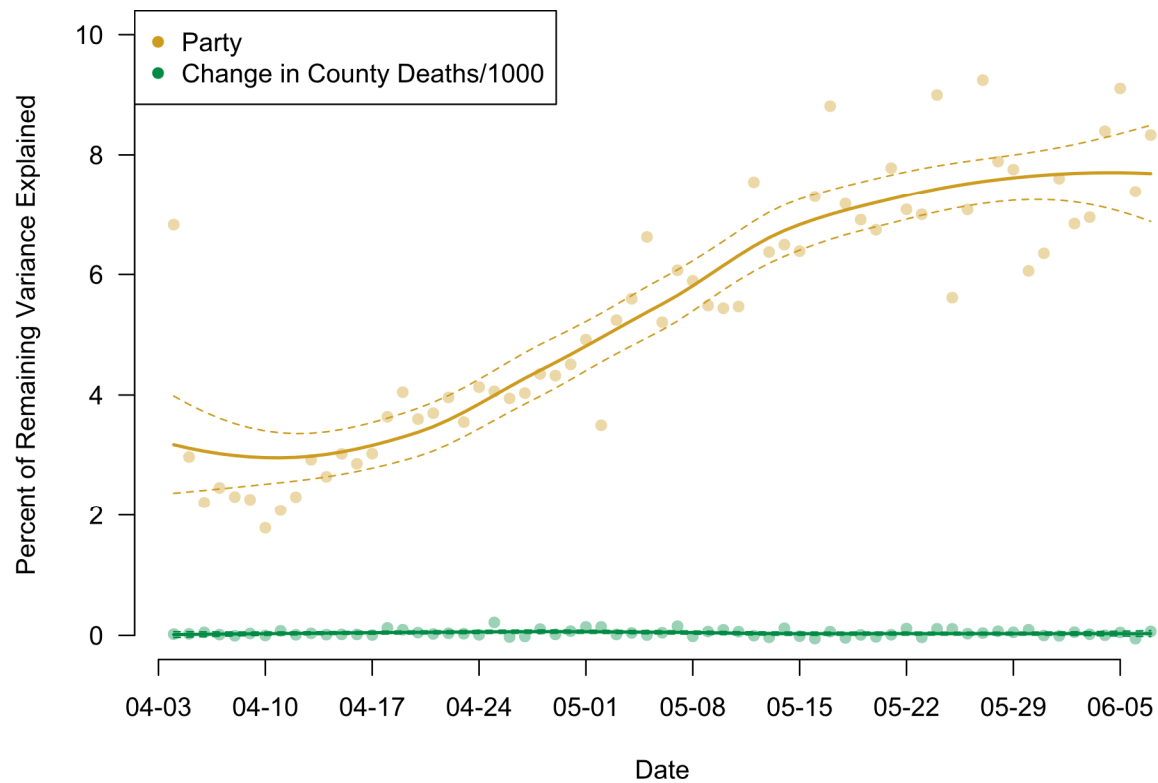


Fig. S6. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (Change in County Deaths/1000). The daily coefficients of partial determinations are computed via OLS regression using specifications (1-3) run separately on each day. Observations weighted to the US adult population. They describe the percent of residual variation in specification (1) that is explained by our two key variables. Trend is local regression.

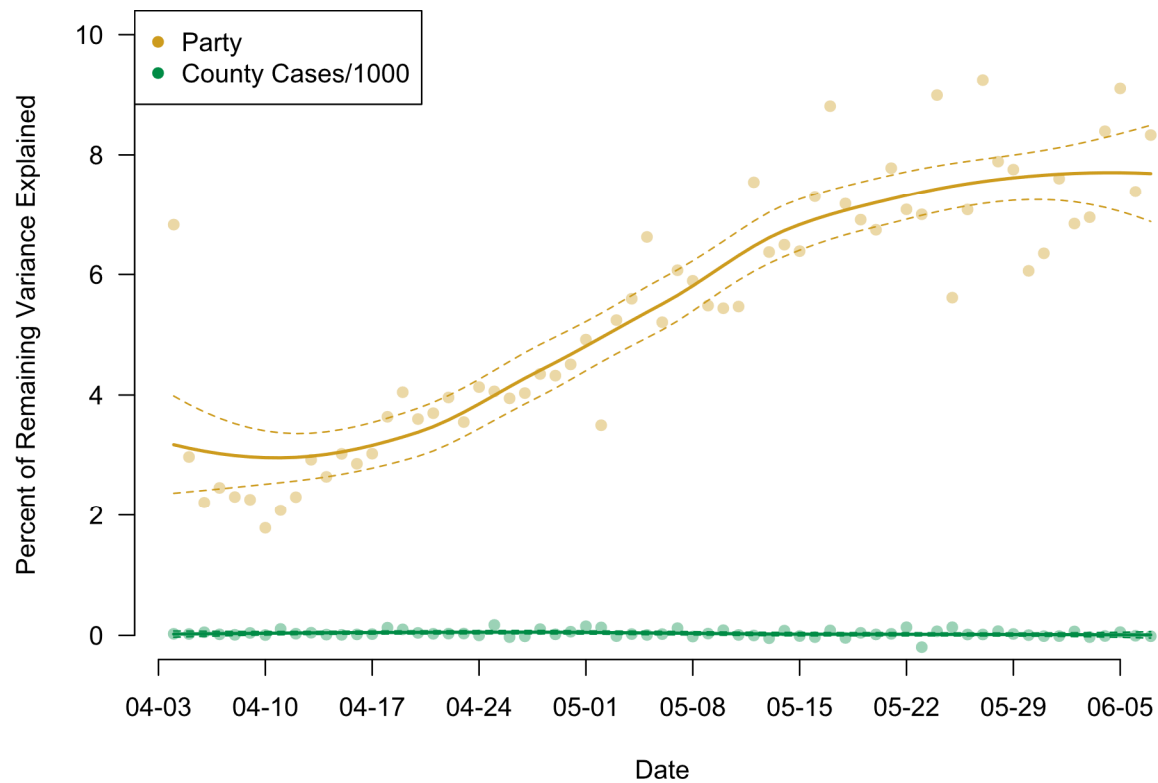


Fig. S7. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (County Cases/1000). The daily coefficients of partial determinations are computed via OLS regression using specifications (1-3) run separately on each day. Observations weighted to the US adult population. They describe the percent of residual variation in specification (1) that is explained by our two key variables. Trend is local regression.

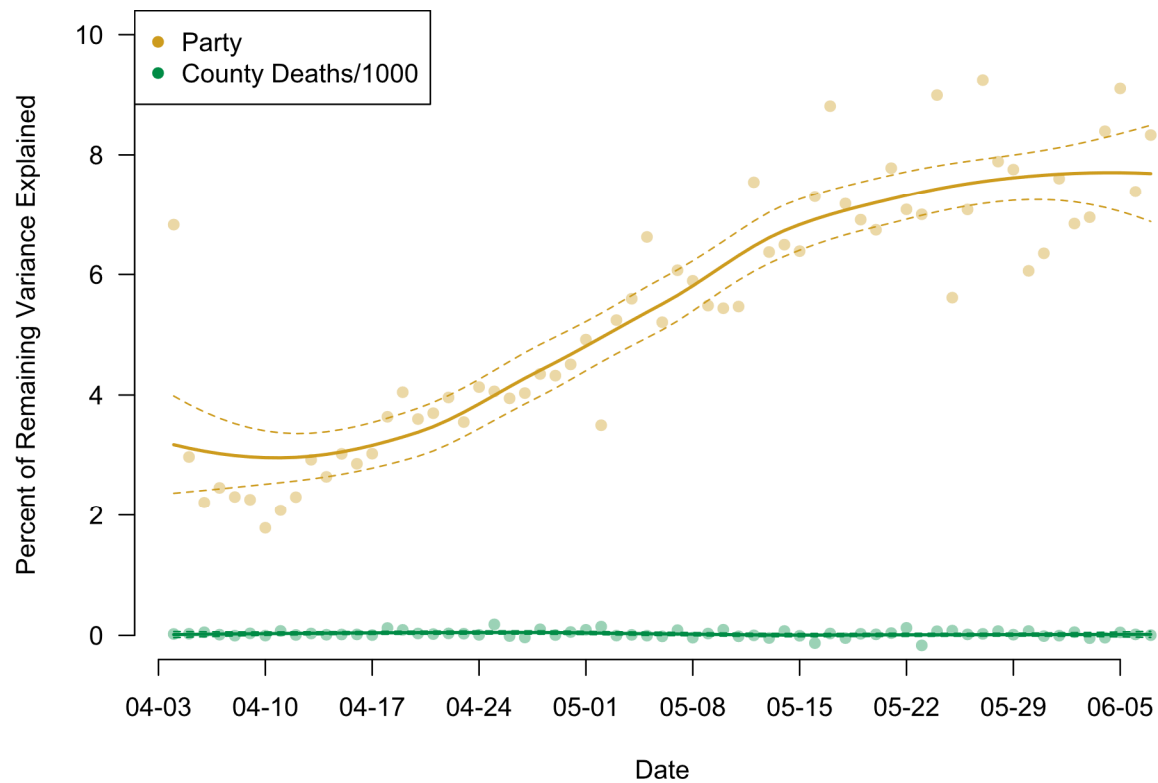


Fig. S8. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (County Deaths/1000). The daily coefficients of partial determinations are computed via OLS regression using specifications (1-3) run separately on each day. Observations weighted to the US adult population. They describe the percent of residual variation in specification (1) that is explained by our two key variables. Trend is local regression

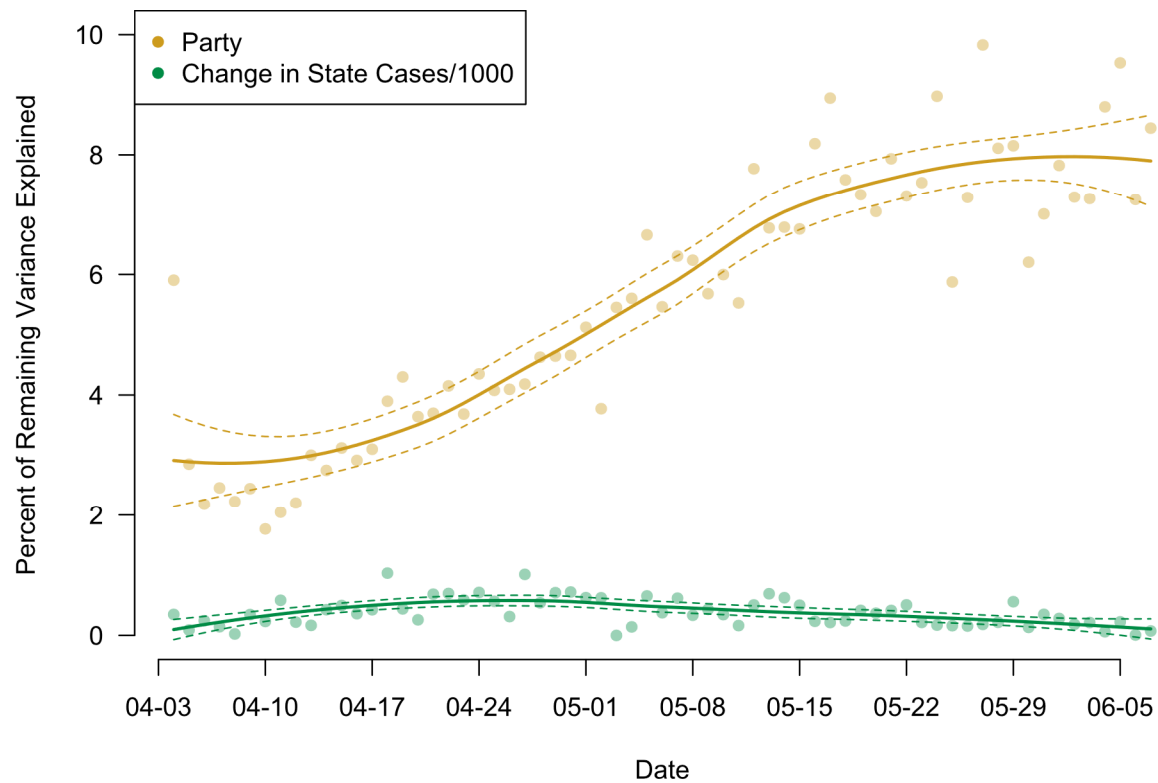


Fig. S9. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (Change in State Cases/1000). The daily coefficients of partial determinations are computed via OLS regression using specifications (1-3) run separately on each day. Observations weighted to the US adult population. They describe the percent of residual variation in specification (1) that is explained by our two key variables. Trend is local regression.

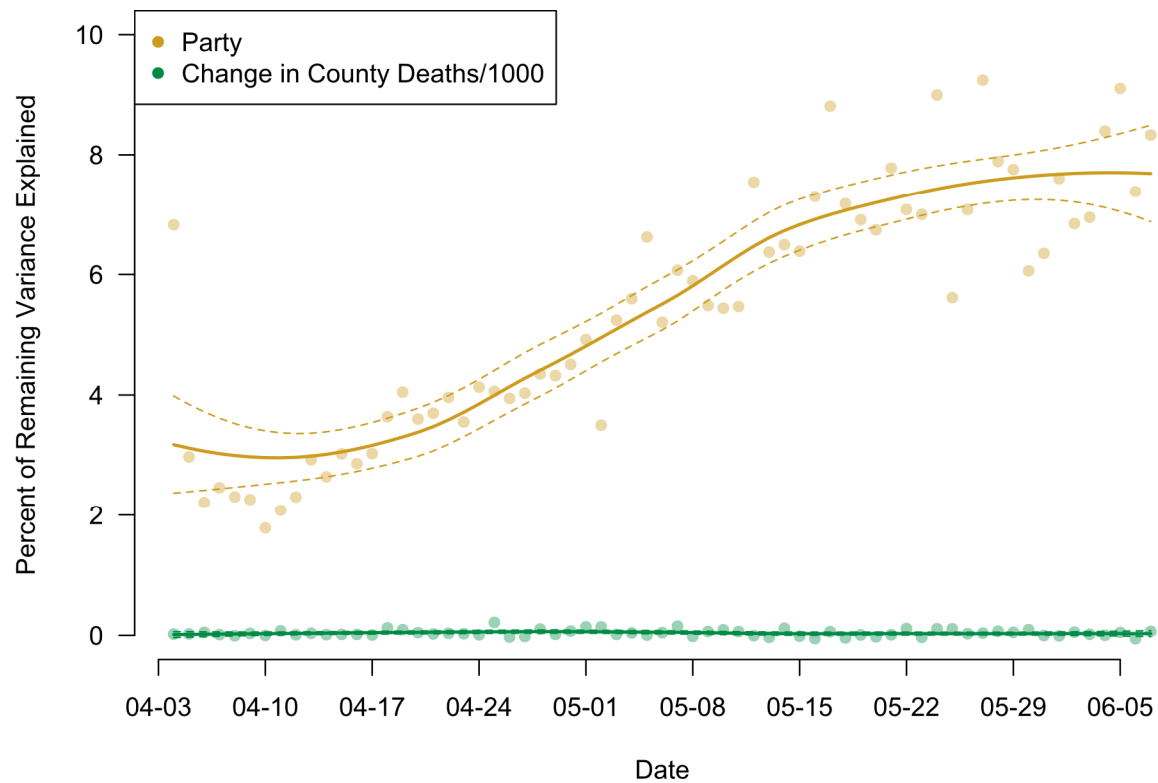


Fig. S10. Variation in Social Distancing Behavior Explained by Partisanship and the Impact of COVID-19 in the Community (Change in State Deaths/1000). The daily coefficients of partial determinations are computed via OLS regression using specifications (1-3) run separately on each day. Observations weighted to the US adult population. They describe the percent of residual variation in specification (1) that is explained by our two key variables. Trend is local regression.

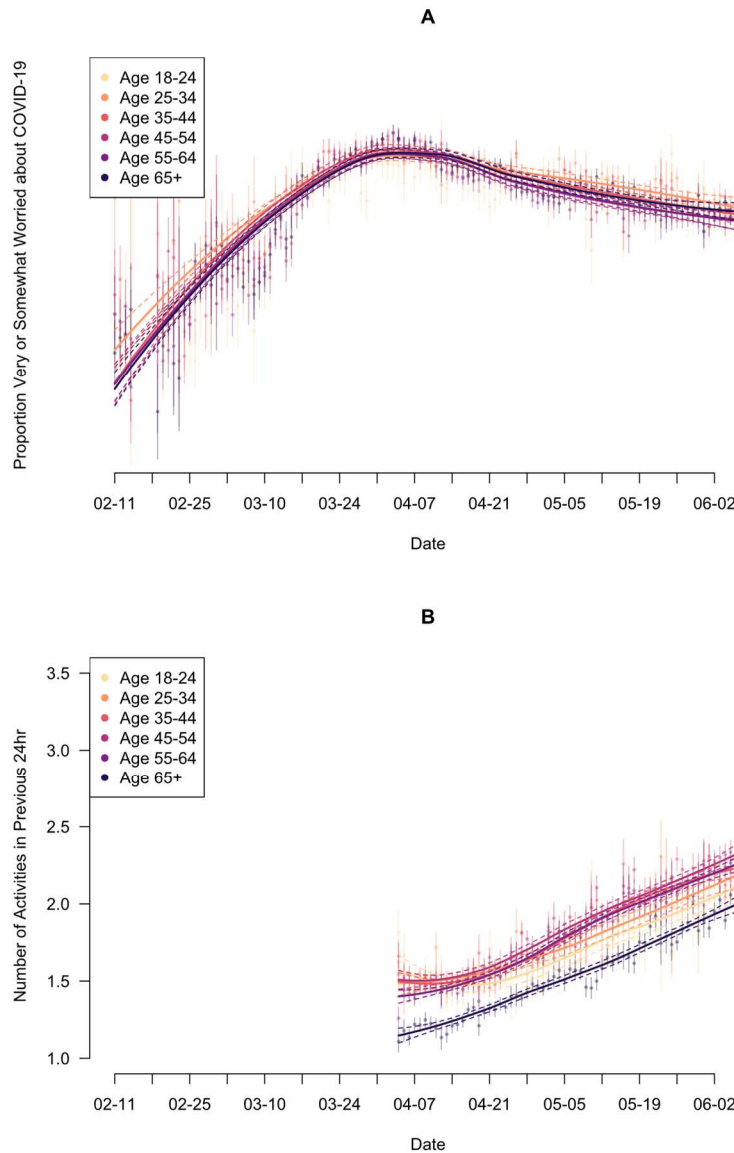


Fig. S11. Concern with Catching COVID-19 and Social Distancing Behavior Over time by Age Group. Plot (A) graphs the weighted daily percentage of respondents who are “very” or “somewhat” worried about catching COVID-19 and Plot B graph the average number of social activities individuals report doing in the last 24 hours. 95% confidence intervals are reported for each daily average and plotted lines are loess smoothers to summarize trends over time.

	Last 24 Hour Activity [0-7]
Time	0.01* (0.001)
Time ²	0.0001* (0.0000)
Democrat	-0.21* (0.01)
Lean Democrat	-0.23* (0.01)
Lean Republican	0.42* (0.01)
Republican	0.37* (0.01)
County Change in Covid Cases/1000	-0.005 (0.01)
Female	-0.35* (0.01)
Income 30k-74K	0.04* (0.01)
Income 75k-149k	0.12* (0.01)
Income 150k+	0.22* (0.02)
Age 25-34	-0.07* (0.01)
Age 34-44	-0.11* (0.02)
Age 45-54	-0.15* (0.02)
Age 55-64	-0.20* (0.02)
Age 65+	-0.27* (0.02)
Race Other	0.04* (0.01)
Black	0.003 (0.01)
Asian	-0.39* (0.02)
Hispanic	-0.13* (0.02)
High School	0.03 (0.02)
Associates/Some College	0.09* (0.03)
College	0.10* (0.03)
Postgraduate	0.07* (0.02)
Unemployed	-0.33* (0.01)
Rural	0.03* (0.01)
Urban	-0.04* (0.01)
State F.E.	Yes
N	425969
Adj. R-squared	0.14
Residual Std. Error	1.29 (df = 425891)

*p < .05

Table S1. Number of Social Activities Regression Results with Leaners. Specification analyzes data pooled over time to allow social distancing to vary between “leaners” and “partisans.” The similarity of coefficients reveals the lack of difference and the justification for collapsing. Standard errors are clustered by state.